

HIROSHIMA, Japan

WMO: 477650

Lat: 34.3983N

Lon: 132.4628E

Elev: 177

StdP: 14.60

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.6	32.2	17.1	13.1	42.8	19.9	15.0	42.4	21.5	44.7	19.3	44.6	7.8	10	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	93.1	77.3	91.4	77.1	89.6	76.4	79.2	88.5	78.3	87.3	77.7	86.4	9.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	140.1	83.8	75.9	136.4	83.4	75.2	132.9	83.0	42.8	88.8	42.0	87.7	41.3	86.6	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.6	15.8	DB	27.3	96.2	2.0	1.7	25.9	97.4	24.8	98.4	23.7	99.4	22.2	100.6
			WB	25.4	80.3	2.3	1.0	23.8	81.0	22.4	81.6	21.2	82.2	19.5	82.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.2	42.1	43.6	49.2	58.6	67.5	74.0	81.5	83.9	77.0	66.6	55.6	46.1	(6)
(7)		DBStd	15.10	4.01	4.77	5.34	5.43	3.93	3.59	4.04	3.18	4.69	5.06	5.53	4.89	(7)
(8)		HDD50	674	248	189	80	5	0	0	0	0	0	13	139	(8)	
(9)		HDD65	2930	711	600	489	200	21	0	0	0	40	283	586	(9)	
(10)		CDD50	5140	2	8	56	262	541	719	975	1051	810	516	182	18	(10)
(11)		CDD65	1923	0	0	0	7	97	269	510	586	361	91	2	0	(11)
(12)		CDH74	17881	0	0	0	29	414	1539	5492	7083	2961	363	0	0	(12)
(13)		CDH80	6512	0	0	0	1	28	264	2144	3154	890	31	0	0	(13)
(14)	Wind	WSAvg	8.0	7.8	8.0	8.5	8.2	7.6	6.9	7.2	7.6	8.5	9.1	8.7	8.2	(14)
(15)	Precipitation	PrecAvg	62.7	1.9	2.5	4.4	6.1	6.2	10.2	9.4	4.8	7.1	4.1	2.8	1.7	(15)
(16)		PrecMax	87.7	5.4	6.2	9.4	10.9	15.7	25.9	20.2	18.2	17.5	15.3	7.9	4.2	(16)
(17)		PrecMin	43.9	0.2	0.4	0.8	2.0	1.6	2.2	0.6	0.2	0.5	0.9	0.1	0.1	(17)
(18)		PrecStd	10.4	1.1	1.3	2.2	2.2	3.0	4.6	5.8	4.0	3.9	2.8	1.9	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	62.1	68.4	77.7	83.0	87.7	95.0	95.8	91.7	83.3	72.6	62.1	(19)
(20)		0.4%	MCWB	47.4	53.7	54.9	61.2	64.2	74.1	77.6	77.0	75.4	71.3	62.1	52.0	(20)
(21)		2%	DB	53.9	57.9	65.0	73.5	80.2	84.9	92.4	93.9	89.3	80.1	70.0	59.3	(21)
(22)		2%	MCWB	45.8	49.3	53.4	58.7	63.6	71.5	77.6	77.1	74.6	67.8	59.9	51.2	(22)
(23)		5%	DB	51.7	54.8	61.9	71.0	78.2	83.0	90.8	92.4	87.2	77.9	67.5	57.0	(23)
(24)		5%	MCWB	44.0	46.5	51.7	57.7	63.0	70.4	77.2	77.1	74.0	66.0	58.4	49.2	(24)
(25)		10%	DB	49.7	52.1	59.1	68.4	76.3	81.2	88.9	90.7	85.1	75.7	65.2	54.6	(25)
(26)		10%	MCWB	42.4	44.5	50.3	56.5	62.3	69.5	76.6	76.9	73.0	64.8	57.1	47.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	56.3	59.2	66.4	70.7	76.5	80.0	80.8	78.6	74.2	65.6	56.1	(27)
(28)		0.4%	MCDWB	55.3	61.0	63.5	70.8	75.4	83.9	90.3	90.8	86.5	80.0	68.5	58.8	(28)
(29)		2%	WB	47.5	51.4	56.3	63.5	68.6	75.0	79.0	79.6	77.3	71.8	63.1	52.6	(29)
(30)		2%	MCDWB	51.5	55.8	61.0	67.9	73.4	81.3	89.1	88.9	84.7	77.5	67.5	57.0	(30)
(31)		5%	WB	45.1	48.3	53.9	61.2	66.9	73.8	78.2	78.8	76.3	68.8	60.8	50.5	(31)
(32)		5%	MCDWB	50.0	52.8	59.9	67.3	72.6	79.4	87.7	87.6	83.6	75.1	65.4	55.6	(32)
(33)		10%	WB	43.3	45.7	51.5	59.3	65.3	72.5	77.4	78.1	75.3	66.5	58.3	48.4	(33)
(34)		10%	MCDWB	48.5	50.9	58.4	65.9	72.1	78.0	86.2	86.8	82.6	73.0	63.8	53.7	(34)
(35)	Mean Daily Temperature Range		MDBR	13.0	13.8	15.4	15.9	15.4	12.1	11.1	12.4	13.0	14.6	14.3	13.3	(35)
(36)		5% DB	MCDBR	15.9	17.0	18.9	18.9	18.4	15.1	13.7	14.6	14.7	16.3	16.0	16.1	(36)
(37)		5% DB	MCWBR	10.0	10.7	11.3	9.7	8.2	6.1	4.6	4.4	5.5	7.6	9.2	10.0	(37)
(38)		5% WB	MCDBR	14.2	14.2	15.3	13.4	12.3	9.8	12.5	12.4	11.7	12.4	12.8	13.9	(38)
(39)		5% WB	MCWBR	10.4	10.8	11.7	8.8	7.5	4.8	4.6	4.5	5.1	7.5	9.2	10.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.412	0.456	0.520	0.549	0.548	0.566	0.566	0.534	0.491	0.445	0.422	0.405	(40)	
(41)		taud	2.098	2.009	1.862	1.847	1.909	1.942	1.996	2.085	2.162	2.193	2.159	2.134	(41)	
(42)		Ebn at Noon	241	244	239	240	242	237	236	241	246	246	237	235	(42)	
(43)		Edh at Noon	40	49	61	65	62	60	56	51	45	40	37	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	690	899	1226	1512	1683	1469	1539	1597	1282	1071	783	607	(44)	
(45)		RadStd	50	75	100	137	167	152	208	155	127	104	62	42	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-1.08	N/A	-0.67	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page